Chapter 6_ Reasoning From fast to WISE

It found the fastest way, but it did not reason through the implications of its decisions. That is the difference between an AI that can think and one that can only react.

The future of AI is not about speed; it is about the ability to think deeply and reason carefully, much like humans handle complex decisions.

For the reasoning model, instead of being trained like normal LLMs to predict the probability of the next word, it is trained to emphasize strong, deliberate reasoning and iterative problem-solving. They are developed using reinforcement learning techniques that encourage the model to refine its thinking, attempt different approaches, recognize errors, and improve its responses over time.

The more time an AI spends reasoning, retrieving, and refining its output, the smarter it gets. That said, an AI agent would not only retrieve information faster but also potentially become more intelligent the longer it works on a task.

Limitations

"Premature closure" = jumping to conclusions before fully processing all the constraints. just like a student rushes into answering the question before fully understanding the question.

"Context fragmentation" = The LLMs process information through their attention mechanisms, which, while powerful, don't truly replicate the human working memory.

"False confidence syndrome" = The model would provide incorrect words with the same high confidence as correct ones, failing to distinguish between different levels of certainty.

The model lacks the type of working memory and constraint satisfaction capabilities that humans use for complex puzzles.

- **Pattern matching challenge:** LLMs don't think the way we do; instead, they perform what is commonly called "next-token-prediction", especially guessing what word or symbol should come next based on patterns they have learned during the training. They were fundamentally trying to match patterns rather than truly reason about the relationships between the different parts of the puzzle.

Thus, instant pattern-matching can be dangerous in any complex decision. The best executive knows when to slow down and reason through implications systematically.

So, we need to build an AI system that gathers and analyzes contextual information before taking actions. This is not about collecting data; it is about creating a comprehensive understanding of the decision environment. In a business context, investing in a robust data infrastructure that enables AI agents to access and process comprehensive contextual information. It also means developing clear protocols for information gathering and validation before critical decisions are made

We'd better give the LLMs more time to analyze and initialize the scope of the project or problem before attempting solutions. We will get better results from the agent when we help them understand the complete context of the request.

Multiple-Agentic AI

Better reasoning often comes not from giving a single AI agent more time but from enabling multiple AI agents to reason together.
In AI terms, a multi-agent system consists of multiple AI agents working together, each potentially having different capabilities, training, or specialized knowledge.

Multiple AI work together > One big AI

Some agents can provide quick, pattern-based responses while others engage in deeper reasoning, much like how humans balance intuitive and analytical thinking in group decision-making.
They are not just about combining strength; they are about creating something entirely new.

For instance, one agent can excel at pattern recognition while another is better at logical deduction. When working together, they create what researchers call "emergent capability," the ability that does not exist in any individual agent but emerges from their interaction.

As a result, multi-agent systems were more effective at predicting and mitigating disruptions compared to their single-agent counterpart.